GLOBAL ACADEMY OF TECHNOLOGY

Department of Aeronautical Engineering • Vimaan Club

VAIMAANIK 2026

Aero Precision RC Mission

RC PLANE COMPETITION | RULEBOOK

12-13 November 2026
National Aero Symposium
Ideal Homes Township, Raja Rajeshwari Nagar, Bengaluru - 560098

Innovate, Inspire and Soar.

DOCUMENT STATUS	ORGANIZER REVIEW DRAFT
Confirmed basis	Two 30-second flight rounds; team-supplied payload; maximum wingspan 1,500 mm; maximum aircraft mass 2 kg without payload.
Pending decisions	Interpretation of the 1:1 thrust-to-weight requirement and the proposed operating/scoring rules. See Section 9 before publication.

1. Event objective

Demonstrate reliable aircraft control through a basic flight round, followed by a payload-carrying flight and a controlled release into a designated drop area. Teams select a payload suited to their aircraft and must complete a proper takeoff and controlled landing.

2. Aircraft limits and definitions

PARAMETER	REQUIREMENT
Aircraft type	Radio-controlled, fixed-wing aircraft.
Maximum wingspan	1,500 mm, measured tip to tip in flight configuration.
Maximum mass without payload	2,000 g, flight-ready: includes battery, electronics, landing gear (if fitted), and the complete payload-release mechanism.
Thrust-to-weight ratio	1:1 without payload. Exact ratio versus minimum, and measurement tolerance, remain to be confirmed.
Payload	Brought by the team; selected to suit the aircraft. A securely closed plastic water bottle is one acceptable proposed option.

The 2 kg limit applies to the aircraft **without payload**. Loaded takeoff mass equals aircraft mass plus payload mass; record both during inspection. No numerical loaded-mass limit has yet been specified by the organizer.

COMPETITION FORMAT

Flight rounds

3. Round 1 - Basic flight

3.1 Present the aircraft without the competition payload and obtain clearance from the flight marshal.

3.2 Perform a proper takeoff from the designated launch/takeoff area. The allowed launch method must be announced before the event.

3.3 Maintain stable, controlled flight for at least **30 continuous seconds**. Timing begins when the aircraft becomes airborne. The aircraft must remain within the approved flight boundary.

3.4 After completing 30 seconds, approach and make a controlled landing in the designated landing area.

3.5 - Proposed qualification Round 1 is a pass/fail qualifying round. All three requirements - successful takeoff, at least 30 seconds of stable flight and a controlled landing - must be completed to enter Round 2.

4. Round 2 - Payload flight and release

4.1 The team brings its own payload. Before flight, judges weigh and record the payload and the complete loaded aircraft. The team may select the payload mass to suit its aircraft, subject to technical inspection.

4.2 Secure the payload so that it cannot shift or detach accidentally. Demonstrate the release mechanism on the ground with the propulsion system made safe.

4.3 Take off with the declared payload attached. Maintain stable flight with the full payload for at least **30 continuous seconds** from becoming airborne.

4.4 After the judge confirms 30 seconds and the drop area is clear, release the payload over the designated area. No release is permitted over spectators, the pits or any other occupied area.

4.5 Maintain control after release and complete a controlled landing in the designated landing area.

4.6 - Proposed scoring condition Payload points count only when takeoff, the 30-second loaded flight, an eligible drop and controlled landing are all successfully completed. A premature or accidental release makes the attempt invalid.

5. Common flight definitions - proposed

5.1 Stable flight: continuous controlled flight, including normal turns, without a stall, spin, uncontrolled descent or ground contact. Aerobatics are not required.

5.2 Controlled landing: the aircraft touches down under control in the designated area, with its main structure intact. Judges distinguish a landing from a crash; minor propeller damage alone does not invalidate a landing.

5.3 Attempt: begins at takeoff/launch and ends at first subsequent ground contact. A go-around without ground contact is allowed; flight time cannot be added across separate launches.

PROPOSED RULES FOR ORGANIZER APPROVAL

Scoring and field procedure

6. Payload-based scoring - proposed

Round 1 determines qualification. Final ranking is based on the valid Round 2 payload score. The following formula makes the points directly proportional to payload mass:

SCORE	CALCULATION
Round 2 payload points	Verified payload mass in grams ÷ 10
Valid drop	The payload's first ground-contact point is inside or on the boundary of the marked drop area. A bounce or roll into the area does not count.
Invalid attempt	0 points: flight shorter than 30 seconds, early release, drop outside the target, no release, or unsuccessful landing.

SUCCESSFULLY DELIVERED PAYLOAD	POINTS
250 g	25
500 g	50
750 g	75
1,000 g	100

These examples are not required payload weights. Weigh the complete released package, including its bottle/container, before flight. Fixed mounting and release hardware remain part of aircraft mass and do not score. Proposed: one payload package and one release per attempt.

6.1 Ranking: highest valid payload score wins. A zero-score attempt does not count as a successful payload delivery.

6.2 Tie-break: smaller distance from the payload's first impact point to the marked target centre wins. If still tied, declare a joint rank unless a common tie-break flight was announced before competition.

7. Attempts and inspection - proposed

7.1 One scoring attempt per round. A reflight is granted only for a verified organizer-caused interruption or an official weather stoppage, with the same opportunity offered consistently to affected teams.

7.2 Before flight, inspect wingspan, flight-ready aircraft mass, loaded mass, structure, control directions, radio range/failsafe, battery condition, payload retention and release operation.

7.3 Verify static thrust with the actual motor, propeller and battery combination in a secured organizer-supervised test. Compute T/W using thrust in newtons divided by aircraft weight in newtons, or thrust in gram-force divided by aircraft mass in grams. Do not apply a pass/fail threshold until the 1:1 rule is finalized.

7.4 Any change to the propulsion combination, airframe or release system requires reinspection. Payload mass changes require reweighing before the attempt.

7.5 The briefing must identify the flight boundary, pilot position, takeoff/landing area, drop-area dimensions, release altitude limits and emergency procedures. Judges record flight time, payload mass, drop result and landing result.

EVENT ADMINISTRATION

Safety, contacts and final decisions

8. Safety and conduct - proposed

8.1 Arm propulsion only in the authorized test/launch area on the marshal's instruction. Keep propellers clear of people; use a working throttle-cut/arming control and a tested radio-loss failsafe.

8.2 Use undamaged batteries, propellers and structural components. Damaged equipment must be repaired or replaced and reinspected before any further flight.

8.3 Only approved, non-hazardous payloads may be carried. No glass containers, sharp exposed parts, flammable materials or loose metal weights. Water bottles must be securely capped and inspected.

8.4 Keep spectators and personnel outside the flight and drop zones. Retrieve the aircraft or payload only after marshal clearance. A pilot must obey a stop-flight or land instruction immediately.

8.5 The marshal may suspend flying for weather, people entering the area or unsafe aircraft behavior. Deliberate unsafe flying, unauthorized drops or refusal to follow safety instructions may result in disqualification.

9. Decisions required before participant publication

This draft preserves the organizer's stated aircraft limits and two-round task. The following details are not yet confirmed:

ITEM	DECISION TO FINALIZE
Thrust-to-weight	Exactly 1:1 or at least 1:1; measurement tolerance and test protocol.
Scoring and attempts	Approve Sections 3.5 and 4.6, and proposed Sections 5-8; confirm qualification, points, tie-break and number of attempts.
Aircraft and field	Hand launch and/or ground takeoff; belly and/or wheeled landing; permitted propulsion/stabilization; safe loaded-mass limit; target size and flight/release limits.
Registration and schedule	Eligibility, team size, fee, registration deadline, RC-event reporting time/date, flying venue and event-specific coordinator.

Brochure alignment: the supplied materials describe organizer-supplied payloads of 0.7 kg and 1 kg and a timed landing mission. This draft follows the organizer's newer instructions: team-supplied payload, two 30-second rounds and payload release. Update the promotional description when the final rules are approved.

10. General event contacts

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Prepared from the supplied VAIMAANIK poster and brochures, the sample RC rulebook structure, and the organizer's updated instructions. The symposium dates are 12-13 November 2026; the RC event's individual slot is to be announced.